

# AIR-1 Air Quality Sensor

*Compact, expandable, locally controlled air-quality monitoring for Home Assistant*

The Apollo AIR-1 is a compact, feature-rich, and expandable air-quality sensor built for Home Assistant. Powered over USB-C and running ESPHome firmware, it is controlled entirely on your local network with no cloud connection or subscription required. The device connects over Wi-Fi and integrates with Home Assistant as well as platforms such as Homey, Hubitat, and HomeSeer.

AIR-1 ships in a finished case measuring just 61 x 61 x 30 mm and includes onboard RGB pixels for color-coded notifications and a power/sensor expansion connector for future add-ons. All firmware and CAD files are open source.

## Key Specifications

|                    |                                                        |
|--------------------|--------------------------------------------------------|
| Product            | AIR-1 Air Quality Sensor                               |
| Manufacturer       | Apollo Automation (Versailles, KY, USA)                |
| Microcontroller    | ESP32-C3-Mini (Espressif)                              |
| Firmware           | ESPHome — open source                                  |
| Connectivity       | Wi-Fi (2.4 GHz) + Bluetooth                            |
| Power input        | USB-C, standard 5 V wall adapter                       |
| Dimensions         | 61 x 61 x 30 mm                                        |
| Onboard indicators | Addressable RGB pixels (color-coded alerts)            |
| Expansion          | Power & sensor expansion connector; GPIO header option |
| Local control      | Yes — no cloud or subscription required                |
| Compatibility      | Home Assistant, Homey, Hubitat, HomeSeer, and more     |
| In the box         | Assembled sensor with case; Apollo sticker             |

## Onboard & Optional Sensors

| Sensor                                   | Measurements                                                                                                                              | Availability |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>SEN55</b><br><i>Sensirion</i>         | PM1, PM2.5, PM4, PM10, VOC index, NOx index, temperature, humidity                                                                        | Standard     |
| <b>SCD40</b><br><i>Sensirion</i>         | CO <sub>2</sub> (carbon dioxide) — pre-calibrated                                                                                         | Optional     |
| <b>MiCS-4514</b><br><i>DFRobot</i>       | CO, ethanol (C <sub>2</sub> H <sub>5</sub> OH), H <sub>2</sub> , NO <sub>2</sub> , ammonia (NH <sub>3</sub> ), methane (CH <sub>4</sub> ) | Optional     |
| <b>ESP32-C3-Mini</b><br><i>Espressif</i> | Bluetooth device / beacon tracking (YAML config required)                                                                                 | Standard     |

The SEN55 particulate sensor is rated for a ~10-year continuous-use lifespan (most comparable PM sensors are rated ~3 years), allowing very short update intervals for applications such as monitoring 3D printers, triggering air purifiers, and detecting paint or solvent fumes.

## Configurations & Pricing

| Configuration                                | CO2 (SCD40) | Gas (MiCS-4514) | Price (USD) |
|----------------------------------------------|-------------|-----------------|-------------|
| Base — particulate, VOC, NOx, temp, humidity | —           | —               | \$109.99    |

| Configuration    | CO2 (SCD40) | Gas (MiCS-4514) | Price (USD) |
|------------------|-------------|-----------------|-------------|
| Base + CO2       | Yes         | —               | \$129.99    |
| Base + CO2 + Gas | Yes         | Yes             | \$199.99    |

*Note: The MiCS-4514 gas sensor has become difficult to source; gas-equipped stock and pricing may be intermittent.*

## Features & Use Cases

- **Particulate monitoring (PM1–PM10):** track dust, pollen, and smoke; automate HVAC fans or air purifiers when air quality declines.
- **CO<sub>2</sub> awareness (optional):** detect elevated indoor CO<sub>2</sub> that can affect sleep, focus, and comfort; trigger ventilation.
- **Gas detection (optional):** monitor CO, ethanol, hydrogen, NO<sub>2</sub>, ammonia, and methane for presence-based automations.
- **Bluetooth tracking:** locate phones, tags, or beacons by proximity to the sensor (YAML configuration required).
- **RGB notifications:** color-coded alerts — e.g. red for poor air quality, green for trash night, blue for mail.
- **Expansion connector:** supports future sensor and power add-on boards (in development).

## Important Notes

- **Temperature offset:** heat from the ESP32-C3 introduces an offset of roughly 2–6 °C; the offset is significantly lower when the sleep function is used.
- **Gas readings:** the MiCS-4514 methane reading can spike when exposed to high levels of other gases. It is best treated as presence/absence detection rather than an accurate PPM readout; it reads ~0 ppm in normal air.
- **Not a safety device:** the AIR-1 does not meet requirements for gas or CO<sub>2</sub> safety monitoring per NBIC, NFPA, or IFC. Do not use it as a life-safety alarm.

## Documentation & Resources

- **Product page:** [apolloautomation.com/products/air-1](https://apolloautomation.com/products/air-1)
- **Wiki / setup guide:** [wiki.apolloautomation.com/products/air1](https://wiki.apolloautomation.com/products/air1)
- **Source & CAD files:** [github.com/ApolloAutomation/AIR-1](https://github.com/ApolloAutomation/AIR-1)
- **Community:** [Apollo Automation Discord](#)
- **Sensor datasheets:** Sensirion SEN55, Sensirion SCD40, DFRobot MiCS-4514, Espressif ESP32-C3-Mini.